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INTRODUCTION

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1.Car and garage paradox : The lack of a notion of absolute simultaneity in special relativity leads to many supposed paradoxes. One of the most famous of these involves a car and a garage of equal proper length. The driver speeds toward the garage, and a doorman at the garage is instructed to slam the door shut as soon as the back end of the car enters the garage. According to the doorman, "the car Lorentz contracted and easily fitted into the garage when I slammed the door." According to the driver, "the garage Lorentz contracted and was too small for the car when I entered the garage." Draw a spacetime diagram showing the above events and explains what really happens. Is the doorman´s statement correct? Is the driver´s statement correct? For definiteness, assume that car crashes through the back wall of the garage without stopping o slowing down.

Solution :

We adopt the Minkowski metric signature $(+,-,-,-)$. he spacetime position four-vector is defined as $X^{\mu} = (ct, x, y, z)$.

Definition of Events in the Garage Frame (S):
Consider a garage of proper length L at rest. A car of proper length L moves at velocity v, relative to the garage, where $\beta = v/c$ and the Lorentz factor is $\gamma = (1-\beta)^{-\frac{1}{2}}$.

Event A (Front Collision): The front of the car reaches the back wall of the garage.

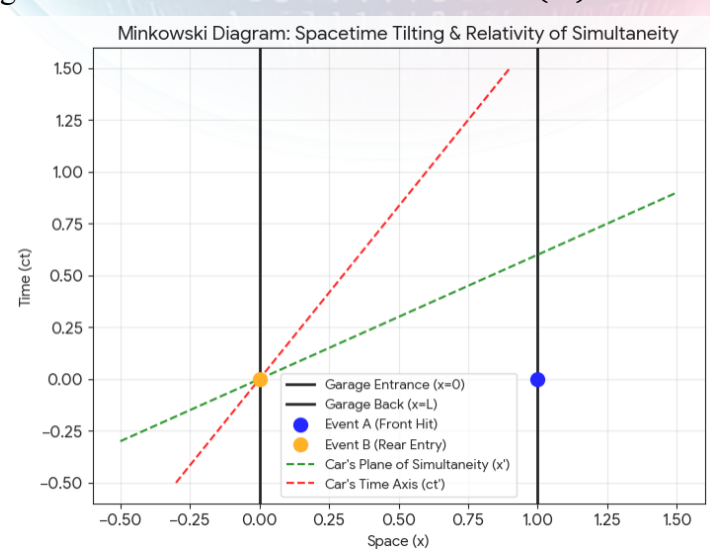
$$X_A = \begin{pmatrix} 0 \\ L \\ 0 \\ 0 \end{pmatrix},$$

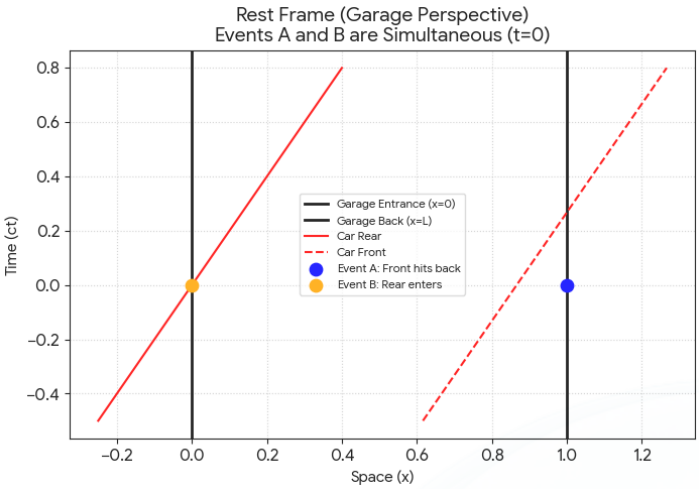
Event B (Rear Entry): The rear of the car passes the garage entrance.

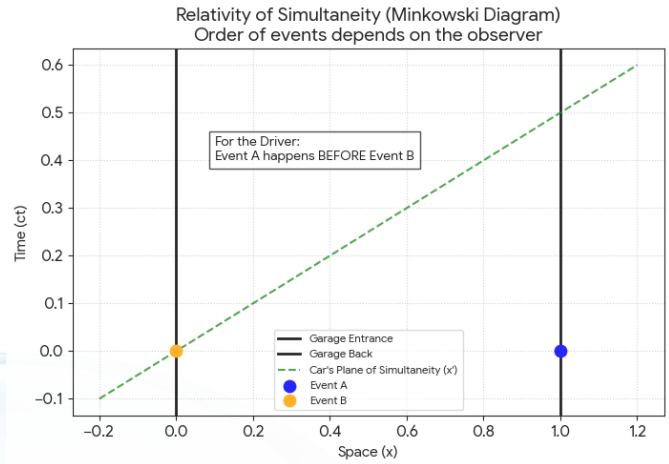
$$X_B = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix},$$

In frame S, these events are defined as simultaneous $(t_A = t_B = 0)$.

The following diagram illustrates the "tilt" of the car's axes (S') due to the Lorentz Boost.







We apply the Lorentz transformation matrix Λ^{μ}_{ν} to map coordinates from the garage frame (S) to the car frame (S').
Temporal Transformation:

$$\Delta ct' = \gamma(\Delta ct - \beta \Delta \mathrm{x}),$$

For events A and B:
Time Interval in S':

$$\Delta ct' = ct'_A - ct'_B = \gamma(0 - \beta L) = -\gamma\beta L,$$

Since $\Delta ct' \neq 0$, simultaneity is destroyed. The negative sign confirms that Event A occurs before Event B in the driver's frame.

Invariance Proof (s^2):

Garage Frame: $s^2 = (c\Delta t)^2 - (\Delta x)^2 = 0^2 - L^2 = -L^2$,
Car Frame: $s^2 = (-\gamma\beta L)^2 - (-\gamma L)^2 = L^2\gamma^2(\beta^2 - 1) = -L^2$,
(Since $\gamma^2(1-\beta^2) = 1$*).*

The interval s^2 is an invariant scalar. While time and space measurements are relative, the spacetime distance between the events is an absolute physical reality.

To provide a complete physical resolution, we must move beyond kinematics and invoke the Conservation of the Energy-Momentum Tensor:

Modeling the car as a relativistic fluid with proper density ρ_0 and four-velocity $u^{\mu} = \gamma(c, v, 0{,}0)$, the tensor components are:

$$T^{\mu\nu} = \rho_0 u^{\mu} u^{\nu} = \gamma^2 \rho_0 \begin{pmatrix} c^2 & cv \\ vc & v^2 \end{pmatrix},$$

$T^{01} = \gamma^2 \rho_0 cv$: Represents the momentum density of the car.

Vanishing Rigidity: In GR/SR, a "rigid body" implies an infinite speed of sound ($v_s > c$), which is a physical impossibility.
Upon Event A (collision), a discontinuity is introduced into the tensor. The conservation law $\partial_\mu T^{\mu\nu} = 0$ requires that:

$$\frac{\partial T^{01}}{\partial(ct)} + \frac{\partial T^{11}}{\partial x} = 0,$$

This differential equation governs the stress wave propagating from the front to the rear. Because the signal is bounded by c the rear momentum (T^{01}) remains constant for $\Delta t \geq \frac{L}{c}$. The car "fits" not just because of a coordinate shift, but because it undergoes physical longitudinal compression induced by the staggered arrival of the stopping signal.

The "Garage Paradox" is not a logical contradiction, but a profound demonstration of the Relativity of Simultaneity and the absolute nature of the Minkowski Interval. The answer to the problem lies in the reconciliation of two seemingly incompatible perspectives:

1. For the Garage Observer (The Doorman): The car physically fits. Due to Lorentz Contraction, the car's length is measured to be less than the garage's length $\left(\frac{L}{\gamma} < L\right)$. Consequently, there is a discrete interval of time where both the front and rear of the car are simultaneously inside the garage.
2. For the Car Observer (The Driver): The garage is contracted $\left(\frac{L}{\gamma} < L\right)$, making it seemingly impossible for the car to fit. However, the Relativity of Simultaneity dictates that the events "front hitting the back wall" and "rear entering the garage" are no longer simultaneous. The driver observes the front of the car colliding with the back wall before the rear has even reached the entrance.

Conclusion:
The paradox is resolved by acknowledging that perfect rigidity is impossible in Special Relativity. Upon impact (Event A), a signal must travel from the front to the rear to "tell" the rest of the car to stop. Since this signal is bounded by the speed of light (c), the rear of the car continues to move forward for a finite time $\Delta t'$, effectively compressing the car.

Final Answer: Both observers are correct. The car "fits" in the garage frame due to geometric contraction, and it "fits" in the car frame due to mechanical compression triggered by the staggered timing of the collision. The invariance of the spacetime interval (s^2) ensures that despite these differing descriptions, they both describe the same objective physical reality: the car ends up inside the garage.

The car crashes at different times for different observers, but it always crashes in a way that respects the Speed of Light. The paradox is solved by realizing that space and time are flexible, but Causality and the Minkowski Interval are absolute.

Colab link for Git Hub animation of Minkowski diagramed :
https://colab.research.google.com/drive/12XsQWas3OTESlvNFZcnyjUWLPGccZi5H?usp=sharing